26900406 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCPHY301 - INTRODUCTION TO QUANTUM MECHANICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

1. State the normalization condition for a wave function and explain its physical meaning. [U] / [CO1]
2. According to Born's statistical interpretation, what does $|\psi(x, t)|^2$ represent? [U] / [CO1]
3. Define probability interference mathematically for electrons and describe how the total probability curve differs from simple classical addition [U] / [CO1]
4. What is an "ideal experiment" according to Richard Feynman? Give an example of an event. [U] / [CO1]
5. For a particle with energy $E > V_0$ approaching a step potential, write the mathematical expressions for the wave numbers k (in the incident region) and l (in the transmitted region). [A] / [CO2]
6. Define the operators a_+ and a_- . Why is the ordering of these operators important? [U] / [CO2]
7. Determine whether the vectors $(1, 2, 0), (0, 1, 1), (1, 3, 1)$ are linearly independent. Justify your answer. [A] / [CO3]

8. Show that $\sin(x)$ and $\sin(2x)$ are orthogonal on $[-\pi, \pi]$. [A] / [CO3]

9. Write down the three properties of eigenvalues and eigenvectors under a Hermitian transformation. [U] / [CO3]

10. Calculate the minimum uncertainty in velocity of an electron if it is confined within a region of width 1 A° [A] / [CO4]

11. Why must a wave function be normalized? [U] / [CO4]

12. Show that the eigen functions of position operator $\hat{x}$ are not in Hilbert space. [U] / [CO4]

[3x10 = 30]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **8** marks

13. Explain how Newton's second law in classical mechanics is analogous to the Schrödinger equation in quantum mechanics in explaining the time evolution of a physical system. [U] / [CO1]

14. Describe the double-slit experiment for classical particles (bullets) and waves. With the help of neat diagrams, explain the observed distribution patterns for single-slit and double-slit conditions. [U] / [CO1]

15. Evaluate the normalization constant for the wave function ψ=Asin(2πx/a) [A] / [CO2]

16. Solve the time-independent Schrödinger equation for a free particle in one dimension. Write down the general solution and discuss the significance of the energy eigenvalues. [U] / [CO2]

17. Define the Hamiltonian operator. Derive the time-independent Schrödinger equation in terms of the Hamiltonian operator [U] / [CO2]

18. Let $T = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ [A] / [CO3]

    (a) Find all the eigenvalues of T by setting up its characteristic equation.

    (b) Determine the eigenvectors corresponding to each eigenvalues.

19. (a) Prove that the product of two unitary matrices is unitary. Is the sum of two unitary matrices also unitary? Explain. [A] / [CO3]

    (b) Is the product of two Hermitian matrices necessarily Hermitian? Explain.

20. Show that $\exp(-x^2/2)$ is an eigenfunction of the operator $\hat{Q} = (d^2/dx^2) - x^2$ and determine the corresponding eigenvalue. [A] / [CO4]

[8x5 = 40]